

3(a)	for a body in (rotational) equilibrium	B1
	sum/total of clockwise moments about a point = sum/total of anticlockwise moments about the (same) point	B1
3(b)(i)	$(W \times 0.45)$ or (19×1.3) or $(W \times 1.85)$ or (22×2.6)	C1
	$(W \times 0.45) + (19 \times 1.3) + (W \times 1.85) = (22 \times 2.6)$ so $W = 14 \text{ N}$	A1
3(b)(ii)	magnitude = $19 + 14 + 14 - 22$ = 25 N	A1
	direction: vertically upwards	A1
3(c)(i)	the extension is zero when the force is zero	B1
	graph is a straight line and (so) Hooke's law obeyed	B1
3(c)(ii)	$k = F/x$ or $k = \text{gradient}$	C1
	e.g. $k = 60 / (1.00 - 0.25)$ $k = 80 \text{ N m}^{-1}$	A1
3(c)(iii)	area shaded below graph line between $L = 0.25 \text{ m}$ and $L = 0.75 \text{ m}$	B1